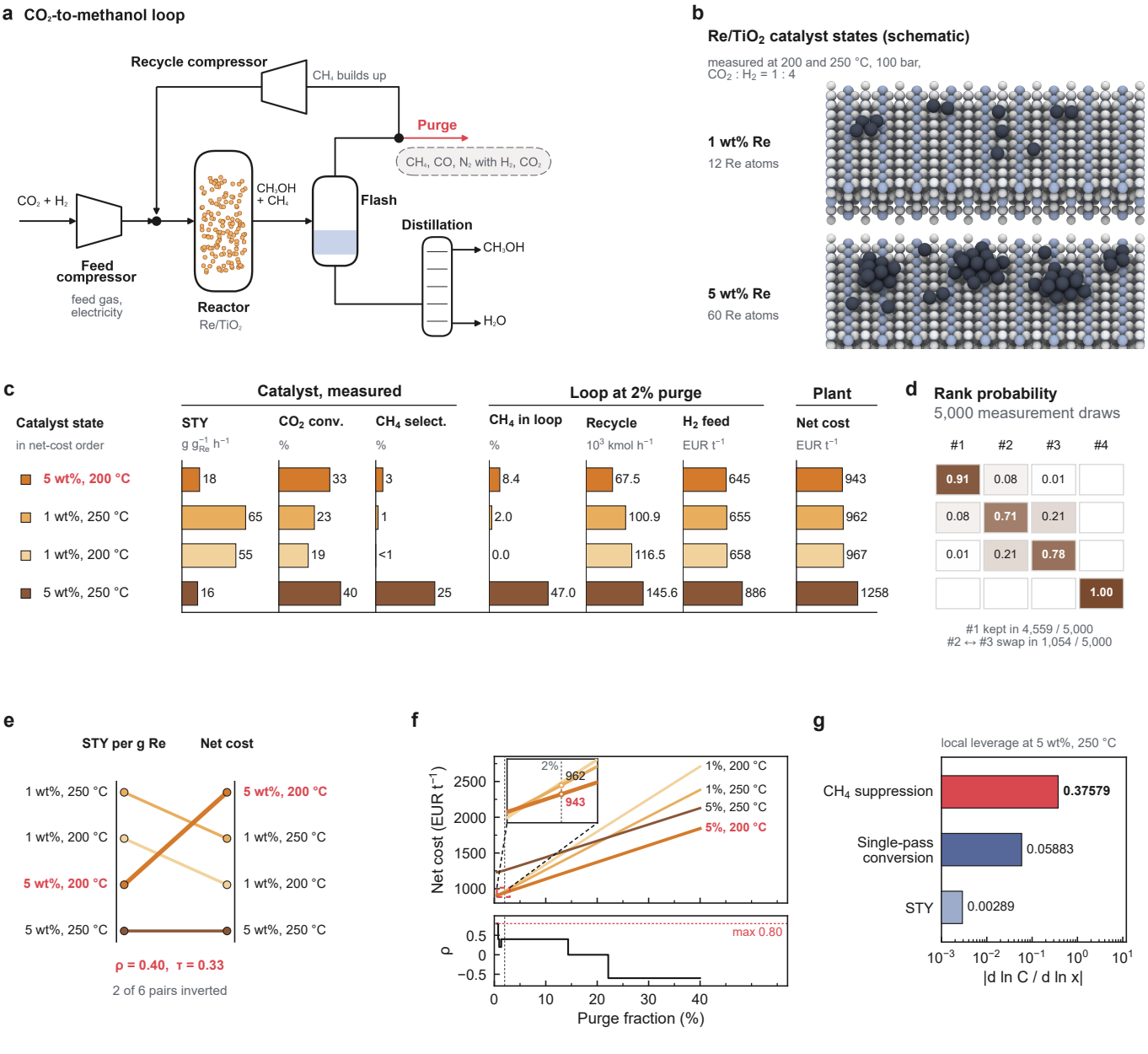

a CO₂-to-methanol loop
Recycle compressor
CH₄ builds up
Purge
CH₄, CO, N₂ with H₂, CO₂
CO₂ + H₂
CH₃OH + CH₄
Flash
Distillation
CH₃OH
H₂O
Feed compressor
feed gas, electricity
Reactor
Re/TiO₂
b
Re/TiO₂ catalyst states (schematic)
measured at 200 and 250 °C, 100 bar, CO₂ : H₂ = 1 : 4
1 wt% Re
12 Re atoms
5 wt% Re
60 Re atoms
c
Catalyst, measured
Loop at 2% purge
Plant
Catalyst state
in net-cost order
STY
g g_Re^-1 h^-1
CO₂ conv.
%
CH₄ select.
%
CH₄ in loop
%
Recycle
10³ kmol h⁻¹
H₂ feed
EUR t⁻¹
Net cost
EUR t⁻¹
5 wt%, 200 °C
18
33
3
8.4
67.5
645
943
1 wt%, 250 °C
65
23
1
2.0
100.9
655
962
1 wt%, 200 °C
55
19
<1
0.0
116.5
658
967
5 wt%, 250 °C
16
40
25
47.0
145.6
886
1258
d
Rank probability
5,000 measurement draws
#1
#2
#3
#4
0.91
0.08
0.01
0.08
0.71
0.21
0.01
0.21
0.78
1.00
#1 kept in 4,559 / 5,000
#2 ↔ #3 swap in 1,054 / 5,000
e
STY per g Re
Net cost
1 wt%, 250 °C
1 wt%, 200 °C
5 wt%, 200 °C
5 wt%, 250 °C
5 wt%, 200 °C
1 wt%, 250 °C
1 wt%, 200 °C
5 wt%, 250 °C
ρ = 0.40, τ = 0.33
2 of 6 pairs inverted
f
Net cost (EUR t⁻¹)
2500
2000
1500
1000
2%
962
943
1%, 200 °C
1%, 250 °C
5%, 250 °C
5%, 200 °C
ρ
0.5
0
−0.5
max 0.80
0
10
20
30
40
Purge fraction (%)
g
local leverage at 5 wt%, 250 °C
CH₄ suppression
0.37579
Single-pass conversion
0.05883
STY
0.00289
10^-3
10^-2
10^-1
10^0
10^1
|d ln C / d ln x|